Pairs Trading with Mean-Reverting Stochastic Processes

Ornstein–Uhlenbeck modelling, ADF-based cointegration selection and optimal trading bands

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1 Detect
Can the ADF and Engle–Granger tests find spreads that really mean-revert?

2 Model
How well does an Ornstein–Uhlenbeck model describe them, and can it be estimated?

3 Trade
Do OU-based rules beat a simple distance rule after costs, in the US and India?

Speakers: Lakshya Nitin Tanishk Vismay Saatwik (3:30 each, 17:30 in total)

The question and the data

Slide 2 • M2 • 1:00 • do two related stocks drift apart for good, or does their gap close?

The idea

- two related stocks, such as KO and PEP, tend to move together
- when the gap opens, buy the cheap one and sell the expensive one
- it pays only if the gap **really closes**

The spread

$$X_t = \ln P_t^A - \beta \ln P_t^B - \mu$$

β, μ from a regression on the formation year only

	US	India (NSE)
Stocks	56 large caps	26
Sectors	7	5
Period	2010–2026	2012–2026
Candidate pairs per window	196	55
Cost per trade, per stock	10 bp	15 bp

Daily adjusted closes from Yahoo Finance; pairs only within a sector.

Today's large caps give the sample **survivorship bias**, so any edge we find would be overstated.

Engle–Granger: testing a pair needs a stricter threshold

Slide 3 • M2 • 1:00 • E2b • 1,000 pairs of independent random walks, never cointegrated

Two steps

1. regress $a_t = \mu + \beta b_t + u_t$ on the formation year; the residual $\hat{u}_t$ is the estimated spread
2. run the ADF test on $\hat{u}_t$

Why the threshold changes

- OLS picks $\hat{\beta}$ to make $\hat{u}_t$ look as stable as possible, so even unrelated prices can pass: a *spurious regression*
- the critical value must be stricter: about -3.34 at 5%, instead of -2.86
- regressing a on b and b on a can disagree, so a pair must pass **both directions**

E2b: how often a false pair is found

Critical values	Wrongly flagged
Plain ADF	**16.9%**
Engle–Granger	**6.2%**
Correct level	5%

Plain ADF would call **one unrelated pair in six** cointegrated; with Engle–Granger the error is close to the intended 5%.

Do the pairs we select stay pairs?

Slide 4 • M2 • 1:30 • E9 • 259 selected US pairs, tested again on the six months after selection

We keep each pair's hedge ratio from selection and run the ADF test (slide 7) on its spread over the next six months.

9%
still mean-revert

52%
expected if unchanged

- only 24 of 259 pairs pass; had each pair kept the speed it showed at selection (median half-life 7 days), the test would flag about half (power from slide 7)
- that benchmark is generous, because fitted speeds are too fast (slide 6); a fairer one is lower, but still well above 9%
- a stronger selection score does not predict a higher profit (correlation about −0.02)

Decile of the selection score (1 = strongest)	1	2	3	4	5	6	7	8	9	10
Mean pair profit per window (%)	1.3	−1.0	−0.2	1.5	−0.1	−1.2	0.5	−0.2	0.3	0.3
Pairs still passing the test (%)	8	8	8	12	8	8	12	12	12	8

Selected pairs **mostly do not keep their mean reversion**, which is why so many of our trades never close (slide 12).

The spread as an Ornstein–Uhlenbeck process

Slide 5 • M3 • 2:00 • one stochastic differential equation, solved exactly with Itô's lemma

The model

$$dX_t = \kappa(\theta - X_t)\,dt + \sigma\,dW_t, \qquad \kappa > 0$$

- κ: speed of the pull back; θ: long-run level; σ: noise
- Itô's lemma on $e^{\kappa t}X_t$ removes the drift:

$$X_t = \theta + (X_0 - \theta)e^{-\kappa t} + \sigma \int_0^t e^{-\kappa(t-s)} dW_s$$

- Gaussian, with variance $\sigma^2(1 - e^{-2\kappa t})/(2\kappa)$
- stationary law $N(\theta, \sigma^2/2\kappa)$; a gap halves in $\ln 2/\kappa$ (the *half-life*)

Exact one-day step $(\Delta = 1/252)$

$$X_{t+\Delta} = a + bX_t + \varepsilon_t, \quad b = e^{-\kappa\Delta}, \quad a = \theta(1 - b)$$

- an AR(1) with Gaussian errors, so **OLS is maximum likelihood**
- no discretisation error (an Euler step would only give $b \approx 1 - \kappa\Delta$)

Estimates from one window

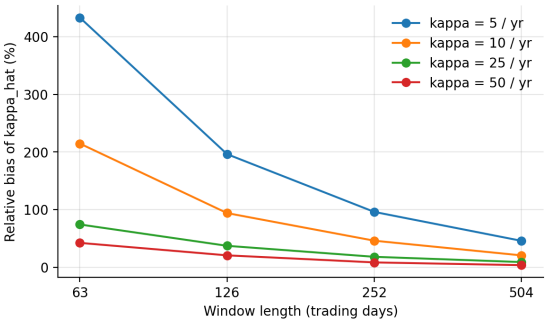
$$\hat{\kappa} = -\frac{\ln \hat{b}}{\Delta}, \quad \hat{\theta} = \frac{\hat{a}}{1 - \hat{b}}, \quad \sigma_{eq} = \frac{\hat{s}_\varepsilon}{\sqrt{1 - \hat{b}^2}}$$

valid only when $0 < \hat{b} < 1$

The discrete step is **exact**, so a regression of tomorrow's spread on today's gives the speed, the level and the noise.

Estimating the speed κ is harder than it looks

Slide 6 • M3 • 1:30 • E1 • 1,000 simulated paths per point, true speed known



With one year (252 days) of data

True speed κ	Half-life	$\hat{\kappa}$ too high by
50	3.5 days	8%
25	7 days	18%
10	17 days	46%
5	35 days	96%

- the bias is always upward, and the error **halves** each time the window doubles
- slower pairs are hit hardest

Pairs look **faster, and more attractive, than they really are**: fitted half-lives are too short, which we see again in E4 and E5.

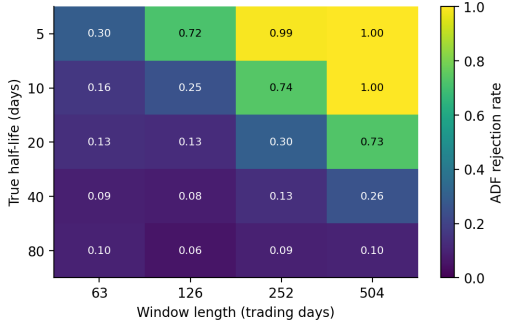
The ADF test asks whether $\kappa = 0$, and one year sees only fast pairs

Slide 7 • M4 • 1:30 • E2a • 300 simulated mean-reverting spreads per cell

Why it tests mean reversion

$$\Delta X_t = a + \gamma X_{t-1} + \sum_{i=1}^{p} \delta_i \, \Delta X_{t-i} + \varepsilon_t, \quad \gamma = e^{-\kappa \Delta} - 1$$

- $H_0 : \gamma = 0$ is exactly $H_0 : \kappa = 0$ (a random walk); $H_1 : \gamma < 0$
- the lagged differences soak up day-to-day correlation in the noise; p is chosen by AIC
- under H_0 the statistic is not Student-t: the Dickey–Fuller 5% critical value is -2.86, not -1.65



How often it finds a gap that really closes

5-day half-life: 99% • 10 days: 74% • 20 days: 30% • 40 to 80 days: 9 to 13% (chance is 5%)

A one-year window reliably detects only **half-lives up to about 10 days**, and every pair we select has a half-life of 21 days or less.

Four trading rules, one protocol

Slide 8 • M4 • 0:45 • thresholds come from the original rules and are never tuned on our results

	Rule	Open a trade when	Close it when
S0	Distance	the gap of normalised prices moves 2σ	the prices cross
S1	Cointegration z-score	$\lvert z\rvert > 2$	$\lvert z\rvert < 0.5$
S2	OU s-score	$s < -1.25$ (long), $s > 1.25$ (short)	$s > -0.50$ (long), $s < 0.75$ (short)
S3	Optimal bands	$\lvert s\rvert \geq m^*$, the best band for the cost (slide 9)	at the opposite band: long flips to short

- S0 picks the pairs with the smallest sum of squared price gaps; S1 to S3 use the selection of slide 3
- $z = X/\sigma_X$ standardises the spread by its formation standard deviation; $s = (X - \hat{\theta})/\sigma_{eq}$ centres it on the fitted OU level
- a position of $+1$ means long \$1 of stock A and short \$$\beta$ of stock B

Same pairs, same costs, same timing for S1 to S3; only the **entry and exit levels** differ.

Bertram's optimal trading bands

One cycle of a trade

enter at a, exit at $m > a$, pay the round-trip cost c; the expected cycle time solves the backward equation $\frac{1}{2}\sigma^2 u'' - \kappa x\, u' = -1$:

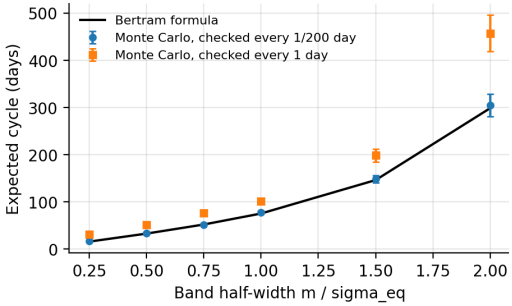
$$\mathbb{E}[T] = \frac{\pi}{\kappa}\left[\operatorname{erfi}\left(\frac{m\sqrt{\kappa}}{\sigma}\right) - \operatorname{erfi}\left(\frac{a\sqrt{\kappa}}{\sigma}\right)\right]$$

$$\mu(a, m) = \frac{m - a - c}{\mathbb{E}[T]} \quad \text{(return per unit time)}$$

- symmetric bands $a = -m$: maximise μ over m to get the optimal band m^*; a higher cost pushes m^* outwards

What E3 shows

- the formula is **exact**: it matches a direct numerical integration to 10^{-8}
- m^*/σ_{eq} depends only on the cost relative to σ_{eq}, not on κ

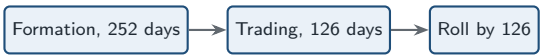


E3: expected cycle time, formula against simulation checked every 1/200 day and daily

Checking the spread once a day makes a cycle **35–99% longer** than the continuous-time formula assumes, because the spread can cross a band and return between two looks.

No look-ahead: how every trade is simulated

Walk-forward protocol



- pairs, β and the OU parameters come from the formation year only and stay frozen
- a signal at the close of day t is traded at the close of $t+1$
- open positions are closed at the end of the window
- costs: 10 bp (US) or 15 bp (India) per stock, per trade

What one trade earns

- a position of $+1$ is long \$1 of A and short \$$\beta$ of B
- daily return on the capital tied up: $(r_t^A - \beta r_t^B)/(1+|\beta|)$, minus the cost of each trade
- returns are measured on all 20 capital slots, idle ones included

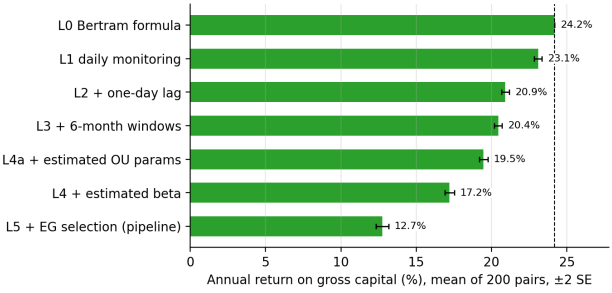
How we know there is no look-ahead

- a test scrambles all prices after a date and checks that no earlier return changes

Letting the parameters peek at the trading window would raise the s-score Sharpe ratio from about **0 to 3.6**: that is why every input is frozen before a trade is placed.

What would a perfect pair earn?

Slide 11 • M1 • 1:00 • E4 • a made-up pair that follows the OU model exactly, run 200 times through the real pipeline



From theory to practice

Annual return, theory (slide 9)	24.2%
Fine-grid simulation	24.3%
Lost to pair selection	−4.5 pts
Daily prices, one-day delay	−3.3
Estimating the hedge ratio	−2.3
Other estimates	−1.0
Six-monthly restarts	−0.5
Full pipeline	**12.7%**

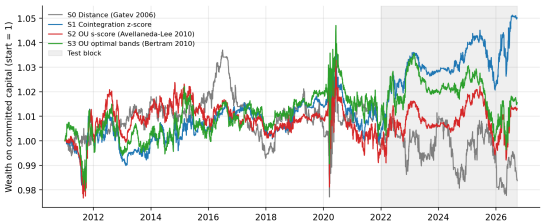
the fine-grid simulation matching the theory shows the profit-and-loss accounting is right

Even a **perfect pair keeps only about half** of its theoretical return, and the biggest single loss is finding it: the selection test picks it in only 74% of windows.

No rule earns a significant return on US stocks, yet the signal is real

Slide 12 • M4 • 1:15 • E5 • 56 stocks, 2011–2026, 10 bp per trade per stock, new pairs every six months

Rule	Return %/yr	Sharpe (SE)	Max DD %
S0 Distance	−0.09	−0.05 (0.25)	−5.8
S1 z-score	0.33	**0.19** (0.25)	−3.9
S2 s-score	0.10	0.06 (0.25)	−4.2
S3 Bertram	0.12	0.06 (0.25)	−4.8



- every Sharpe ratio is within one margin of error of zero; the rules cannot be ranked with confidence
- beta to the market is about 0.02
- **the signal is real:** before costs S1 earns about 44 bp per trade, and S1 to S3 win about two trades in three

The losses come from the **28–45% of trades that never close** within six months (about −3% each); they are common because only about 8 of 196 candidate pairs pass the test, no more than chance.

Does it hold in India?

Slide 13 • M5 • 1:30 • E8 • 26 NSE stocks in 5 sectors, 2013–2026, 15 bp per trade per stock

Rule	Return %/yr	Sharpe (SE)	Trades won	Profit per trade
S0 Distance	0.43	0.16 (0.27)	54%	24 bp
S1 z-score	0.41	**0.38** (0.27)	68%	90 bp
S2 s-score	0.37	0.32 (0.27)	68%	42 bp
S3 Bertram	0.43	0.32 (0.27)	69%	57 bp

- all four rules gain slightly, **none significantly**: the same picture as the US
- in the last test period (2022–2026) S1 to S3 stay positive (Sharpe 0.28 to 0.82) and S0 turns negative; that is only 4.7 years, so the margin of error is about 0.46
- after costs the cointegration rules win about two trades in three

Why so little: few pairs pass

- 55 candidate pairs per window, only about 3 pass the selection test; chance alone gives 2.8
- 4 of 27 windows have no pair at all, so about 3 of the 20 capital slots are used
- 83 pairs selected in 13 years, 42 of them banks

India repeats the US result: a real per-trade signal (up to 90 bp after costs), too few genuine pairs for a significant overall return.

Conclusions

1 Detect

- a year of data finds only half-lives up to about 10 days (E2a)
- Engle–Granger passes about as many pairs as chance (8 of 196)
- most selected pairs soon stop mean-reverting (E9)

2 Model

- the OU model and Bertram's formula are correct (E3)
- but κ is hard to estimate: 18% too high even for a fast pair (E1)
- a perfect pair earns only about half its theory (E4)

3 Trade

- before costs the rules catch real mean reversion (about two trades in three win)
- after costs no rule is significant, in the US or India (E5, E8)
- the weak link is **pair selection**, not the trading rules

Limits survivorship bias, a fixed hedge ratio within each window, daily closes only.
Next a larger universe of stocks, a more stable pair selection, a time-varying hedge ratio.

A real, trade-level signal that selection cannot yet turn into a significant profit: the **bottleneck is choosing genuinely cointegrated pairs**.